

Assignment 4 – FOMC Meetings Analysis using NLP, Problem Statement & Background

The Federal Open Market Committee (FOMC) is a branch of the Federal Reserve System (the Fed), the central banking system of the United States. The FOMC is primarily responsible for overseeing the nation's open market operations (i.e., buying and selling government securities) and setting monetary policy, including influencing interest rates, regulating money supply, and adjusting bank reserve requirements. The committee meets several times a year to discuss the economic and financial conditions of the country, make policy decisions, and set the direction for future monetary policy.

FOMC Meeting Minutes are detailed records of the discussions and decisions made during these meetings. They provide insights into the economic conditions that influenced the committee's conclusions, the members' views on the economic outlook, and their rationale for any changes to monetary policy. These minutes are released three weeks after each meeting. Economists, traders, and financial analysts gauge the direction of future policy and closely analyze them. (<https://www.federalreserve.gov/monetarypolicy/fomccalendars.htm>)

Natural Language Processing (NLP) in Analyzing FOMC Meeting Minutes

NLP can be a powerful tool in extracting useful information from the FOMC meeting minutes.

- **Sentiment Analysis:** NLP can help determine the sentiment of the text, whether it's positive, negative, or neutral. By analyzing the sentiment of the minutes, stakeholders can gauge the committee's confidence in the economy, which can influence financial markets.
- **Topic Modeling:** NLP techniques like Latent Dirichlet Allocation (LDA) or Non-negative Matrix Factorization (NMF) can identify the main topics discussed in the minutes. This helps in understanding the focus areas of the meetings, such as inflation, employment rates, or specific economic sectors.
- **Trend Analysis:** NLP can track the frequency and context of specific terms or phrases over time, indicating shifts in focus or policy stance. For example, increasing mentions of "inflation" might predict tighter monetary policy.
- **Summarization:** Automated summarization tools can provide quick and concise summaries of lengthy documents, saving time and highlighting key points.

Usage in Companies

- Many companies, especially in the financial sector, use NLP to gain an edge in understanding and predicting market dynamics based on the FOMC minutes:
- Financial Analysts and Investors use NLP to quickly gather insights from the minutes, helping them make informed investment decisions.
- Economic Consultancies and research firms use NLP to supplement their analyses and reports, providing clients with detailed interpretations of monetary policy.
- Banks and Financial Institutions leverage NLP to anticipate changes in interest rates and adjust their loan and mortgage strategies accordingly.
- Fintech Companies may develop tools that integrate NLP to offer real-time analytics and alerts based on the release of such minutes.

NLP's ability to process and analyze large volumes of text data makes it invaluable for extracting insights from complex documents like the FOMC meeting minutes, enabling faster and more informed decision-making across various sectors.

Data

1. Please construct a corpus from 2022 – present (March 2025) of FOMC meeting minutes on FOMC monetary policy (Just the meeting minutes)
2. Perform a second data set and follow all the instructions below. Use the guardian widget and create a dataset using the topic of US economy from Jan 1, 2022, to present

Instructions (Separate sections for both datasets) – You are acting as a Monetary policy analyst

1. Please construct a dataset for the meeting minutes and a separate for the guardian scrapping
2. Perform text pre-processing
3. Perform text EDA in detail – use all available widgets, including word cloud, collocation, text statistics, and n-gram analysis – Please explain the observations
4. Perform sentiment analysis and perform EDA on the output – plot sentiment trends over time for both data sets
5. Perform topic analysis and LDA clusters for both data sets– Use a bag of words if you like for your embedding (You can choose the total number of clusters based on your analysis)
6. Perform corpus to network and conduct network analysis using the network explorer for both data sets
7. **Summarize observations between the two datasets.** Do you see the effect of the FOMC monetary policy on the news? – What other observations can you get from this analysis

Submission

Along with PowerPoint, please upload a copy of*.ows file

Naming convention – StudentNameAssignment_#.ppt, StudentNameAssignment_#.ows